No.	Solution	Remarks
3(a)(i)	Since pressure at A is equal to pressure at B, volume is directly proportional to thermodynamics temperature $\frac{V_A}{V_B} = \frac{T_A}{T_B}$ $T_B = T_A \times \frac{V_B}{V_A}$ $= 315 \times \frac{7.6}{2.8}$ (only ratio of $\frac{V_B}{V_A}$ applies) $= 855$ K	[1] for correct expression and numerical substitution [1] for correct answer
3(a)(ii)	Work done = area under the *p*-*V* graph $W = p\Delta V$ $= (3.5 \times 10^5)\left[(7.6 - 2.8) \times 10^3 \times 10^{-6}\right]$ $= 1680$ J	[1] for correct expression and numerical substitution [1] for correct answer
3(b)	(table below) Explanation: From B to C: there is no change in temperature, so there is no change in internal energy, i.e. $\Delta U_1 = 0$ From C to A: there is no change in volume, so there is no work done, i.e. $W = 0$. The change in internal energy, $\Delta U_2 = Q + W = Q = -2520$ J From A to B: let the change in internal energy be ΔU_3 For a complete cycle, B → C → A → B, the total change in internal energy = 0, i.e. $\Delta U_1 + \Delta U_2 + \Delta U_3 = 0$ → $\Delta U_3 = 0 - \Delta U_1 - \Delta U_2$ $= 0 - 0 - (-2520)$ $= 2520$ J	[3] – 1 mark for each correct entry

stage	increase in thermal energy / J
from point B to point C	0
from point C to point A	−2520
from point A to point B	+2520

No.	Solution	Remarks